

AP Statistics FRQ Practice Pack

Regenerated edition - topic coverage retained; layout reorganized by difficulty; author/publisher branding removed.

Study use only. Problem contexts follow publicly released AP Statistics FRQ templates. Practice with hints/templates; do not use on graded work unless allowed.

How this pack is organized

1. Four topics kept intact (Exploring Data → Collecting Data → Probability & RVs → Inference).
2. Within each topic, problems are ordered by difficulty:
 - * Foundation - identify / interpret / short explanation
 - Standard - calculation + justification in context
 - * Challenge - full multi-step inference or layered probability
3. Substantive stems unchanged (cleaned for OCR/layout only).
4. Part II contains reference answers for every problem (worked solutions for self-check).

Topic map (CED alignment)

Topic | CED units | Typical FRQ skills

1 Exploring Data | Units 1-2 | SOCS, outliers, regression interpret, residuals, r^2

2 Collecting Data | Unit 3 | Sampling bias, SRS/stratify, experiments, confounding

3 Probability & RVs | Units 4-5 | Conditional probability, Binomial/Geometric/Normal, EV

4 Statistical Inference | Units 6-9 | CI/tests for p & μ , χ^2 , slope inference

Part I - Practice Problems

Topic 1 Exploring Data

T1-F1 Foundation * Scatterplot language & residual

Template source label (for study tracking): 2017 FRQ1 (wolves)

Researchers studying a pack of gray wolves collected data on length x (meters, nose to tip of tail) and weight y (kilograms). A scatterplot of weight versus length was described as positive, linear, and strong.

(a) Explain what is meant by each of the following for this situation:

(i) Positive

(ii) Linear

(iii) Strong

The least-squares equation is $\hat{y} = -16.46 + 35.02x$.

(b) Interpret the slope of the least-squares regression line in context.

(c) One wolf with length 1.4 m had residual -9.67 kg. What was the weight of the wolf?

T1-F2 Foundation * Two-way table proportions & association

Template source label (for study tracking): 2014 FRQ1 (extracurricular)

An administrator sampled 100 students. Residential status (on/off campus) and extracurricular participation are summarized:

Participation | On campus | Off campus | Total

No activities | 9 | 30 | 39
One activity | 17 | 25 | 42
Two or more | 7 | 12 | 19
Total | 33 | 67 | 100

(a) Calculate the proportion of on-campus students who participate in no extracurricular activity, and the same proportion for off-campus students.

(b) Using the segmented bar graph idea (relative frequencies within each residential status), write a few sentences summarizing the association between residential status and level of participation in the sample.

T1-S1 Standard ** Five-number summary & two outlier rules

Template source label (for study tracking): 2021 FRQ1 (hospital stay)

Length of stay (days) for a random sample of 50 patients:

Days | 5 | 6 | 7 | 8 | 9 | 12 | 21
Patients | 4 | 13 | 14 | 11 | 6 | 1 | 1

(a) Determine the five-number summary of the distribution of length of stay.

(b) Method A = 1.5 x IQR rule; Method B = 2 standard deviations from the mean.

(i) Using Method A, identify potential outliers. Justify.

(ii) Mean = 7.42 days, SD = 2.37 days. Using Method B, identify potential outliers. Justify.

(c) Explain why Method A might identify more outliers than Method B for a distribution that is strongly skewed right.

T1-S2 Standard ** Histogram SOCS, outliers, boxplot vs histogram

Template source label (for study tracking): 2019 FRQ1 (room sizes)

Room sizes (sq ft) for 20 rooms are shown in a histogram (bimodal, roughly 100-350). Summary statistics:

Mean 231.4, SD 68.12, Min 134, Q1 174, Median 253.5, Q3 292, Max 315.

(a) Based on the histogram, write a few sentences describing the distribution of room size.

(b) Determine whether there are potential outliers. Then sketch a boxplot of room size.

(c) What characteristic of the shape is apparent from the histogram but not from the boxplot?

T1-S3 Standard ** Describe distribution; mean vs median under a change

Template source label (for study tracking): 2016 FRQ1 (tips)

A histogram shows Robin's 60 tip amounts for one day (right-skewed, roughly \$0-\$25).

(a) Write a few sentences describing the distribution of tip amounts.

(b) One tip was \$8. If that tip had been \$18 instead, what effect would the change have on the mean and on the median? Justify.

T1-S4 Standard ** Slope, r^2 , largest residual

Template source label (for study tracking): 2022 FRQ1 (bullfrogs)

A biologist fits a least-squares line for predicting bullfrog mass (g) from length (mm):
 $\text{mass} = -546 + 6.08(\text{length})$, with $r^2 \approx 0.819$. (Plot of 11 bullfrogs with the line is given.)

(b) Identify and interpret the slope in context.

(c) Interpret $r^2 \approx 0.819$ in context.

(d) From the residual pattern on the plot:

(i) Approximately what is the length and mass of the bullfrog with the largest absolute residual?

(ii) Does the LSRL overestimate or underestimate that bullfrog's mass? Explain.

T1-S5 Standard ** Intercept, r^2 , outlier on scatterplot

Template source label (for study tracking): 2018 FRQ1 (checkout)

A random sample of 11 customers: x = customers ahead in line, y = checkout time (seconds).

Computer output: Constant 72.95; Customers in line 174.40; $R\text{-Sq} = 73.33\%$; $S = 200.01$.

(a) Identify and interpret the intercept estimate in context.

(b) Identify and interpret r^2 in context.

(c) One point is an outlier. Circle it on the scatterplot and explain why it is an outlier.

Topic 2 Collecting Data

T2-F1 Foundation * Random assignment to two equal groups

Template source label (for study tracking): 2009 FRQ3 (frog labs)

A teacher will compare physical frog dissection vs computer simulation using 24 students. Analyze posttest - pretest score changes.

(a) Describe a method for assigning the 24 students to two groups of equal size that allows a statistically valid comparison of the two programs.

T2-F2 Foundation * Identify treatments, units, response; control; random assignment

Template source label (for study tracking): 2019 FRQ2 (fungus)

Researchers create four fungus concentrations: 0, 1.25, 2.5, 3.75 ml/L. Equal numbers of insects are placed in 20 containers; each container is sprayed with one mixture. Record number still alive after one week.

(a) Identify treatments, experimental units, and response variable.

(b) Does the experiment have a control group? Explain.

(c) Describe how to randomly assign treatments so each treatment has the same number of units.

T2-S1 Standard ** Matched pairs vs CRD

Template source label (for study tracking): 2022 FRQ2 (acne twins)

A dermatologist recruits 36 pairs of identical twins. Each person receives new drug or placebo for 2 weeks; improvement scored 0-100. Twins have similar acne severity; severity differs across pairs.

(a) Identify treatments, experimental units, and response variable.

(b) Describe a statistical advantage of a matched-pairs design (twins paired) rather than a completely randomized design.

T2-S2 Standard ** Median vs mean; voluntary response vs SRS

Template source label (for study tracking): 2014 FRQ4 (alumni income)

At a reunion, mean income of 50 attendees was reported as the class mean income.

(a) Statistical advantage of using the median of reported incomes rather than the mean as a typical-income estimate?

(b) Method 1: email all 6,826 alumni (expect ≥ 600 responses). Method 2: SRS and phone until all selected respond (about 100 possible). Which method for estimating average yearly income of all class members? Compare.

T2-S3 Standard ** Convenience, SRS, stratification

Template source label (for study tracking): 2013 FRQ2 (campus survey)

Estimate proportion of 70,000 students satisfied with campus appearance; sample size 500.

(a) Why might surveying the first 500 students entering a football stadium be biased?

(b) Describe how to select an SRS of 500 from the list of 70,000 names.

(c) When would stratification by campus give a more precise estimate than stratification by gender?

T2-S4 Standard ** Observational study & confounding

Template source label (for study tracking): 2016 FRQ3 (smoking & Alzheimer's)

A study tracked 21,123 people for 23 years. For those smoking ≥ 2 packs/day, Alzheimer's risk was 2.57x that of nonsmokers. Headline: "Strong Association Between Smoking and Alzheimer's."

(a) Identify explanatory and response variables.

(b) Observational study or experiment? Explain.

(c) Explain how exercise status could be a confounding variable.

T2-C1 Challenge *** Experiment scope & non-random assignment

Template source label (for study tracking): 2011B FRQ2 (acrophobia)

27 people: 17 randomly assigned D-cycloserine + 2 therapy sessions; 10 get placebo + 2 sessions. Evaluated after 3 months.

(a) Experiment or observational study? Explain.

(b) Drug group improved significantly more than placebo. Can researchers conclude drug + 2 sessions is as beneficial as 8 sessions without the pill? Justify.

(c) If therapists chose who got the drug (no randomization), why might that lead to an incorrect conclusion?

Topic 3 Probability & Random Variables

T3-S1 Standard ** Joint, union, conditional; independence

Template source label (for study tracking): 2019 FRQ3 (medicine adherence)

Relative frequencies:

| Never | Sometimes | Always | Total
Men | 0.0564 | 0.2016 | 0.2120 | 0.4700
Women | 0.0636 | 0.1384 | 0.3280 | 0.5300
Total | 0.1200 | 0.3400 | 0.5400 | 1.0000

One person selected at random.

- (a)(i) $P(\text{never and woman})$
 - (a)(ii) $P(\text{never or woman})$
 - (a)(iii) $P(\text{never} \mid \text{woman})$
 - (b) Are “never” and “woman” independent? Justify.
-

T3-S2 Standard ** Combinations & flawed simulation

Template source label (for study tracking): 2014 FRQ2 (convention)

6 men and 3 women; 3 selected at random; all 3 selected were women.

- (a) $P(\text{selecting 3 women})$.
 - (b) Based on (a), reason to doubt the manager's “random” claim?
 - (c) Simulation: roll 3 dice; 1-4 = man, 5-6 = woman; count triple-woman trials. Does this correctly simulate selecting 3 women from 9 people (6M, 3W)? Why/why not?
-

T3-S3 Standard ** Discrete RV: P , $E(X)$, conditional

Template source label (for study tracking): 2015 FRQ3 (ATMs)

X = number of working ATMs when the mall opens. $P(X=0)=0.15$, $P(1)=0.21$, $P(2)=0.40$, $P(3)=0.24$.

- (a) $P(\text{at least one working})$
 - (b) $E(X)$
 - (c) $P(\text{all three working} \mid \text{at least one working})$
 - (d) Given at least one working, is $E(X \mid X \geq 1)$ less than, equal to, or greater than $E(X)$ from (b)? Explain.
-

T3-S4 Standard ** Binomial definition, $P(\text{at least one})$, expected value

Template source label (for study tracking): 2021 FRQ3 (gift cards)

200 employees; each week one chosen uniformly at random (with replacement across weeks; independent).

- (a) For a particular employee in 52 weeks:
 - (i) Define the RV and its distribution.
 - (ii) $P(\text{at least one gift card})$.
 - (b) Expected number of gift cards for a particular employee in 52 weeks; interpret.
 - (c) An employee gets none all year - strong evidence selection was not random?
-

T3-S5 Standard ** Geometric wait; binomial count; surprise

Template source label (for study tracking): 2011B FRQ3 (upgrades)

$P(\text{upgrade}) = 0.10$ each flight, independent.

- (a) $P(\text{first upgrade occurs after the third flight})$.
 - (b) $P(\text{exactly 2 upgrades in next 20 flights})$.
 - (c) In 104 flights, would more than 20 upgrades be surprising? Justify.
-

T3-C1 Challenge *** Sequential testing; rare event reasoning

Template source label (for study tracking): 2016 FRQ4 (igniters)

Overall igniter failure rate claimed 15%. Test super igniters until first failure or 32 successes.

- (a) If failure rate is 15%, $P(\text{first 30 succeed})$.
 - (b) Given first 30 succeed, $P(\text{first failure on 31st or 32nd})$.
 - (c) Given first 30 succeed, is it reasonable to believe failure rate $< 15\%$? Explain.
-

T3-C2 Challenge *** Normal underfill + Binomial crate rejection

Template source label (for study tracking): 2022 FRQ3 (shampoo)

$A \sim N(0.60 \text{ L}, 0.04 \text{ L})$. Underfilled if $A < 0.50$. Boxes of 10; crate rejected if a random box has ≥ 2 underfilled.

- (a) $P(\text{a bottle underfilled})$.
 - (b)(i) Define RV for warehouse manager and its distribution.
 - (b)(ii) $P(\text{crate rejected})$.
 - (c) Adjusted program: $N(0.56, 0.03)$. Recommend original or adjusted? Justify.
-

T3-C3 Challenge *** Normal percentile, claim probability, expected gain

Template source label (for study tracking): 2019 FRQ5 (battery warranty)

Battery life $\sim N(30 \text{ months}, 8 \text{ months})$. \$50 warranty replaces phone if life < 24 months.

- (a) In how many months is it expected that 25% of batteries no longer work?
 - (b) $P(\text{a warranty customer requires replacement within 24 months})$.
 - (c) Gain +\$50 if no claim; loss -\$150 if claim. Expected gain per warranty.
-

T3-C4 Challenge *** Law of total probability; Bayes; Binomial

Template source label (for study tracking): 2018 FRQ3 (left-handed)

3.5% of births are multiple births. Of multiple-birth children, 22% left-handed; of single-birth, 11% left-handed.

- (a) $P(\text{randomly selected child is left-handed})$.
 - (b) $P(\text{multiple birth} \mid \text{left-handed})$.
 - (c) For a random sample of 20 children, $P(\text{at least 3 left-handed})$.
-

Topic 4 Statistical Inference

T4-S1 Standard ** One-proportion z-interval + claim about 0.5

Template source label (for study tracking): 2022 FRQ4 (streaming)

Random sample of 920 U.S. teens; 59% use a video streaming service every day.

- (a) Construct and interpret a 95% CI for the true proportion.
 - (b) Based on the CI, do the data provide convincing evidence the true proportion is not 0.5? Justify.
-

T4-S2 Standard ** One-prop CI → cost interval

Template source label (for study tracking): 2017 FRQ2 (water cups)

Of 80 random customers who asked for a water cup, 23 filled it with soda.

- (a) Construct and interpret a 95% CI for the true proportion who do this.
 - (b) Each soda-fill costs \$0.25. If 3,000 ask for water cups in June, use the CI to estimate June's cost.
-

T4-S3 Standard ** CI for regression slope

Template source label (for study tracking): 2019 Intl FRQ2 (house prices)

n=20 houses. Regression of price (thousands \$) on distance (miles): slope $b=-2.158$, $SE_b=0.149$. Conditions met.

- (a) Construct and interpret a 95% CI for the true slope ($t^*\approx 2.101$, $df=18$).
 - (b) Agent believes price drops about \$2,000 per mile (slope -2). Does the CI contradict that belief?
-

T4-S4 Standard ** Proportions vs counts; homogeneity hypotheses

Template source label (for study tracking): 2021 FRQ5 (soft drinks)

Independent city samples (Yes = consumed soft drink in past week):

Baltimore	Detroit	San Diego	Total
Yes 727	1232	1482	3441
No 177	431	798	1406
Total 904	1663	2280	4847

- (a) A researcher claims Baltimore has the lowest likelihood of Yes because 727 is the smallest Yes count. Correct? Explain.
 - (b)(ii) Which city has the smallest Yes proportion? Give the value.
 - (c)(i) Appropriate inference procedure to compare Yes proportions across cities.
 - (c)(ii) Hypotheses.
-

T4-C1 Challenge *** One-proportion z-test + Type I/II

Template source label (for study tracking): 2021 FRQ4 (\$10 coupon)

Manager believes with a \$10 coupon, more than 40% of past customers will reorder. Of 90 randomly selected past customers, 38 reorder. $\alpha=0.05$.

- (a) Is there convincing evidence the manager is correct? Complete the inference procedure.
 - (b) Based on your conclusion, which error (Type I or II) could have been made? Interpret in context.
-

T4-C2 Challenge *** Two-proportion z-test for an increase

Template source label (for study tracking): 2019 FRQ4 (kochia)

2014: 19.7% of 61 plants resistant. 2017: 38.5% of 52 plants resistant. At $\alpha=0.05$, convincing evidence the proportion resistant increased?

T4-C3 Challenge *** Causation from design + two-sample t

Template source label (for study tracking): 2018 FRQ4 (ACL surgery)

210 patients randomly assigned to standard or new ACL procedure.

(a) Would a significant result allow concluding the new procedure causes reduced recovery for patients like these? Explain.

(b) Standard: $n=110$, $\bar{x}=217$, $s=34$. New: $n=100$, $\bar{x}=186$, $s=29$.

Convincing evidence new has smaller mean recovery?

T4-C4 Challenge *** Chi-square association

Template source label (for study tracking): 2013 FRQ4 (fruits/vegetables)

Random sample of 8,866 adults: age group vs whether they eat ≥ 5 servings of fruits/vegetables daily.

Age	Yes	No	Total
18-34	231	741	972
35-54	669	2242	2911
55+	1291	3692	4983
Total	2191	6675	8866

Convincing evidence of association between age-group and 5+ servings/day?

T4-C5 Challenge *** Paired t-test

Template source label (for study tracking): 2014 FRQ5 (car prices)

8 car models: one woman and one man bought the same equipped model. Differences (woman - man): mean $\bar{d} = \$585$, $s_d = \$530.71$.

Do the data provide convincing evidence that, on average, women pay more than men for the same car model?

T4-C6 Challenge *** Chi-square goodness-of-fit

Template source label (for study tracking): 2015 Intl FRQ4 (bank waits)

Acceptable wait-time probabilities: 0.30, 0.25, 0.20, 0.15, 0.10.

Observed counts in $n=100$: 25, 21, 21, 20, 13.

Conduct a test: are waiting times inconsistent with the acceptable probabilities?

Part II - Reference Answers

Use after attempting. Score yourself on: correct method, work shown, and context in the conclusion.

T1-F1 Scatterplot language & residual

Difficulty: Foundation *

- (a) (i) Positive: as length increases, weight tends to increase.
 - (ii) Linear: the pattern of points is well described by a straight line.
 - (iii) Strong: points are tightly clustered about the line (little scatter).
 - (b) For each additional 1 meter of length, the predicted weight increases by about 35.02 kg.
 - (c) $\hat{y} = -16.46 + 35.02(1.4) = 32.568$.
 - Actual $y = \hat{y} + \text{residual} = 32.568 - 9.67 = 22.90$ kg.
-

T1-F2 Two-way table proportions & association

Difficulty: Foundation *

- (a) On campus: $9/33 \approx 0.273$.
 - Off campus: $30/67 \approx 0.448$.
 - (b) Off-campus students have a higher proportion with no activities; on-campus students have higher proportions with one activity and with two or more. Residential status and participation level appear associated in this sample.
-

T1-S1 Five-number summary & two outlier rules

Difficulty: Standard

- (a) Min = 5, Q1 = 6, Median = 7, Q3 = 8, Max = 21.
 - (b)(i) IQR = 2. Fences: $6 - 1.5(2) = 3$ and $8 + 1.5(2) = 11$.
 - Potential outliers: 12 and 21 (both > 11).
 - (b)(ii) Interval: $7.42 \pm 2(2.37) = (2.68, 12.16)$.
 - Potential outlier: 21 only.
 - (c) Right skew pulls the mean and SD upward, pushing the 2-SD upper cutoff farther out. Quartiles/IQR are resistant, so the 1.5xIQR upper fence stays lower and can flag more high values.
-

T1-S2 Histogram SOCS, outliers, boxplot vs histogram

Difficulty: Standard

- (a) The distribution is bimodal (peaks near lower and upper ranges), spanning about 100-350 sq ft, with no extreme outliers obvious from the histogram. Center is near the mid-200s; spread is moderately large.
 - (b) IQR = 118. Fences: $174 - 1.5(118) = -3$ and $292 + 1.5(118) = 469$.
 - Min/Max within fences → no outliers.
 - Boxplot: whiskers 134-315; box Q1=174 to Q3=292; median 253.5.
 - (c) Bimodality (two peaks) - visible in the histogram, not in the boxplot.
-

T1-S3 Describe distribution; mean vs median under a change

Difficulty: Standard

- (a) Tip amounts are strongly right-skewed, centered near about \$5-\$10, ranging from about \$0 to \$25, with a few larger tips in the right tail.
- (b) Mean increases (total increases by \$10).
Median is unchanged ($n = 60$; changing one value that stays on the same side of the center does not move the middle order statistics).
-

T1-S4 Slope, r^2 , largest residual

Difficulty: Standard

- (b) Slope = 6.08: for each additional 1 mm of length, predicted mass increases by about 6.08 g.
- (c) About 81.9% of the variation in bullfrog mass is explained by the linear relationship with length.
- (d)(i) Identify the point farthest vertically from the line (largest $|\text{residual}|$) from the plot.
- (d)(ii) If the point is above the line, the model underestimates; if below, it overestimates. (On the source plot, the largest $|\text{residual}|$ is above the line $\rightarrow$ underestimate.)
-

T1-S5 Intercept, r^2 , outlier on scatterplot

Difficulty: Standard

- (a) Intercept ≈ 72.95 : when there are 0 customers ahead, predicted checkout time is about 72.95 seconds (may be extrapolation).
- (b) $r^2 = 73.33\%$: about 73.33% of the variation in checkout time is explained by the linear relationship with number of customers in line.
- (c) The outlier is the point that does not follow the linear pattern (large residual and/or unusual combination of x and y relative to the others).
-

T2-F1 Random assignment to two equal groups

Difficulty: Foundation *

Label students 1-24. Use a random number generator (or shuffle slips) to select 12 distinct numbers for Group A (physical dissection); the remaining 12 are Group B (simulation). Equal size + chance assignment supports a fair comparison.

T2-F2 Identify treatments, units, response; control; random assignment

Difficulty: Foundation *

- (a) Treatments: 0, 1.25, 2.5, 3.75 ml/L.
Units: the 20 containers.
Response: number of insects alive after one week.
- (b) Yes - 0 ml/L is a control (no fungus).
- (c) Label containers 1-20; randomly permute. Assign first 5 to 0, next 5 to 1.25, next 5 to 2.5, last 5 to 3.75 (5 each).
-

T2-S1 Matched pairs vs CRD

Difficulty: Standard

(a) Treatments: new drug, placebo.

Units: the 72 people (or 36 twin pairs as matched units).

Response: improvement score 0-100.

(b) Pairing controls for genetic/severity similarity within twins, reducing variability and making treatment differences easier to detect than in a CRD.

T2-S2 Median vs mean; voluntary response vs SRS

Difficulty: Standard

(a) Median is resistant to skew/outliers; income is often right-skewed, so the median better represents a “typical” income.

(b) Prefer Method 2. Method 1 is voluntary response/nonresponse biased (e.g., higher earners more likely to reply). Method 2 is smaller but more likely representative/unbiased.

T2-S3 Convenience, SRS, stratification

Difficulty: Standard

(a) Football attendees are not representative of all students; opinions about grounds may correlate with who attends games → bias.

(b) Number students 1-70,000; use RNG to select 500 unique numbers; survey those students.

(c) When satisfaction differs substantially between campuses (and is more homogeneous within campus) than between genders.

T2-S4 Observational study & confounding

Difficulty: Standard

(a) Explanatory: smoking amount (e.g., packs/day). Response: developing Alzheimer’s.

(b) Observational - no treatment assigned; medical histories were tracked.

(c) Exercise may be associated with smoking and with Alzheimer’s risk, so its effect can be mixed with smoking’s effect.

T2-C1 Experiment scope & non-random assignment

Difficulty: Challenge *

(a) Experiment - treatments randomly assigned.

(b) No - the study never compared to an 8-session no-drug group; conclusions are limited to treatments actually compared.

(c) Selection bias/confounding: therapists might assign the drug to milder cases (or vice versa), mixing patient differences with treatment effects.

T3-S1 Joint, union, conditional; independence

Difficulty: Standard

(a)(i) 0.0636

(a)(ii) $0.1200 + 0.5300 - 0.0636 = 0.5864$

(a)(iii) $0.0636 / 0.5300 = 0.12$

(b) Yes - $P(\text{never} | \text{woman}) = 0.12 = P(\text{never})$.

T3-S2 Combinations & flawed simulation

Difficulty: Standard

(a) $C(3,3)/C(9,3) = 1/84 \approx 0.0119$.

(b) Yes - about 1.2% is unusually small if selection were random.

(c) No - dice model allows repeats and treats trials as independent with $P(\text{woman})=1/3$ each roll; true selection is without replacement.

T3-S3 Discrete RV: P, E(X), conditional

Difficulty: Standard

(a) $1 - 0.15 = 0.85$

(b) $0(0.15)+1(0.21)+2(0.40)+3(0.24) = 1.73$

(c) $0.24/0.85 \approx 0.282$

(d) Greater - conditioning removes $X=0$, shifting the mean up (≈ 2.035).

T3-S4 Binomial definition, P(at least one), expected value

Difficulty: Standard

(a)(i) $X = \# \text{ cards for that employee in 52 weeks; } X \sim \text{Bin}(52, 1/200)$.

(a)(ii) $P(X \geq 1) = 1 - (199/200)^{52} \approx 0.229$.

(b) $E(X) = 52/200 = 0.26$: over many years, about 0.26 cards per 52-week year on average.

(c) No - $P(X=0) \approx 0.771$, so getting none is common under randomness.

T3-S5 Geometric wait; binomial count; surprise

Difficulty: Standard

(a) First three not upgraded: $0.9^3 = 0.729$.

(b) $C(20,2)(0.1)^2(0.9)^{18} \approx 0.285$.

(c) $X \sim \text{Bin}(104, 0.1)$, $\mu=10.4$, $\sigma \approx 3.06$. $P(X > 20)$ is tiny (≈ 0.0005) $\rightarrow$ yes, surprised.

T3-C1 Sequential testing; rare event reasoning

Difficulty: Challenge *

(a) $0.85^{30} \approx 0.00763$.

(b) $0.15 + 0.85(0.15) = 0.2775$.

(c) Yes - under 15% failure, $P(\geq 30 \text{ successes})$ is very small (≈ 0.0076), so observing 30 successes

suggests a lower failure rate.

T3-C2 Normal underfill + Binomial crate rejection

Difficulty: Challenge *

(a) $z = (0.50 - 0.60)/0.04 = -2.5 \rightarrow P \approx 0.0062$.

(b)(i) $X = \# \text{ underfilled in the box of } 10; X \sim \text{Bin}(10, p \approx 0.0062)$.

(b)(ii) $P(X \geq 2) = 1 - P(0) - P(1) \approx 0.0017$.

(c) Prefer original - adjusted mean raises underfill p (≈ 0.0228) and rejection probability (≈ 0.021).

T3-C3 Normal percentile, claim probability, expected gain

Difficulty: Challenge *

(a) 25th percentile: $z \approx -0.6745 \rightarrow 30 - 0.6745(8) \approx 24.6$ months.

(b) $P(X < 24) = P(Z < -0.75) \approx 0.2266$.

(c) $E = 50(1 - 0.2266) + (-150)(0.2266) \approx \4.67 .

T3-C4 Law of total probability; Bayes; Binomial

Difficulty: Challenge *

(a) $P(L) = 0.035(0.22) + 0.965(0.11) = 0.11385$.

(b) $P(M | L) = 0.035(0.22)/0.11385 \approx 0.0676$.

(c) $X \sim \text{Bin}(20, 0.11385); P(X \geq 3) \approx 0.402$.

T4-S1 One-proportion z-interval + claim about 0.5

Difficulty: Standard

(a) $\hat{p} = 0.59$. CI: $0.59 \pm 1.96\sqrt{0.59 \cdot 0.41/920} \approx (0.558, 0.622)$.

We are 95% confident the true proportion of U.S. teens who stream daily is between about 0.558 and 0.622.

(b) Yes - 0.5 is not in the interval.

T4-S2 One-prop CI $\rightarrow$ cost interval

Difficulty: Standard

(a) $\hat{p} = 23/80 = 0.2875$. CI $\approx (0.188, 0.387)$.

95% confident the true proportion is between about 0.188 and 0.387.

(b) Cost = $3000 \times 0.25 \times \text{proportion} \rightarrow$ about (\$141, \$290).

T4-S3 CI for regression slope

Difficulty: Standard

(a) $-2.158 \pm 2.101(0.149) \approx (-2.471, -1.845)$.

95% confident the true slope is between about -2.471 and -1.845 (thousands of \$ per mile).

(b) No - -2 is inside the interval.

T4-S4 Proportions vs counts; homogeneity hypotheses

Difficulty: Standard

(a) Incorrect - must compare proportions, not counts. Baltimore's proportion $727/904 \approx 0.804$ is actually the largest.

(b)(ii) San Diego: $1482/2280 = 0.65$.

(c)(i) Chi-square test of homogeneity.

(c)(ii) H_0 : Yes probability is the same in all three city teen populations.

H_a : at least one city's probability differs.

T4-C1 One-proportion z-test + Type I/II

Difficulty: Challenge *

(a) $H_0: p=0.40$ vs $H_a: p>0.40$.

$\hat{p} = 38/90 \approx 0.422$.

$z = (0.422 - 0.40) / \sqrt{0.4 \cdot 0.6 / 90} \approx 0.43$; $p\text{-value} \approx 0.33 > 0.05$.

Fail to reject H_0 - not convincing evidence that more than 40% will order.

(b) Type II possible: failing to reject a false H_0 - concluding the belief isn't supported when the true rate really is $>40\%$, possibly missing a useful promotion.

T4-C2 Two-proportion z-test for an increase

Difficulty: Challenge *

$H_0: p_{2017} = p_{2014}$ vs $H_a: p_{2017} > p_{2014}$.

Using counts $\approx 12/61$ and $20/52$: pooled $\hat{p} \approx 0.283$, $z \approx 2.21$, $p\text{-value} \approx 0.014 < 0.05$.

Yes - convincing evidence of an increase.

T4-C3 Causation from design + two-sample t

Difficulty: Challenge *

(a) Yes - random assignment supports a causal conclusion for patients like those studied.

(b) $H_a: \mu_{\text{new}} < \mu_{\text{std}}$.

$t = (186 - 217) / \sqrt{34^2/110 + 29^2/100} \approx -7.13$; $p \approx 0$.

Yes - convincing evidence of shorter mean recovery for the new procedure.

T4-C4 Chi-square association

Difficulty: Challenge *

Chi-square test of independence/association.

$\chi^2 \approx 8.98$, $df=2$, $p \approx 0.011 < 0.05$.

Yes - convincing evidence of association.

T4-C5 Paired t-test

Difficulty: Challenge *

Paired t-test: $H_0: \mu_d = 0$ vs $H_a: \mu_d > 0$.

$t = 585 / (530.71 / \sqrt{8}) \approx 3.12$; $df = 7$; $p \approx 0.008 < 0.05$.

Yes - convincing evidence women pay more on average.

T4-C6 Chi-square goodness-of-fit

Difficulty: Challenge *

χ^2 GOF: H_0 : proportions equal claimed; H_a : at least one differs.

Expected: 30, 25, 20, 15, 10.

$\chi^2 \approx 4.09$, $df=4$, $p \approx 0.39 > 0.05$.

Fail to reject - data are not inconsistent with the acceptable probabilities.

Suggested practice path

1. Finish all * in Topics 1-2 (speed + template language).
 2. Finish across all topics (exam pacing).
 3. Save * inference/probability for timed sets (25-45 minutes).
 4. Rewrite conclusions from memory without looking at Part II.
-

End of regenerated pack. No author attribution retained from the source booklet cover.